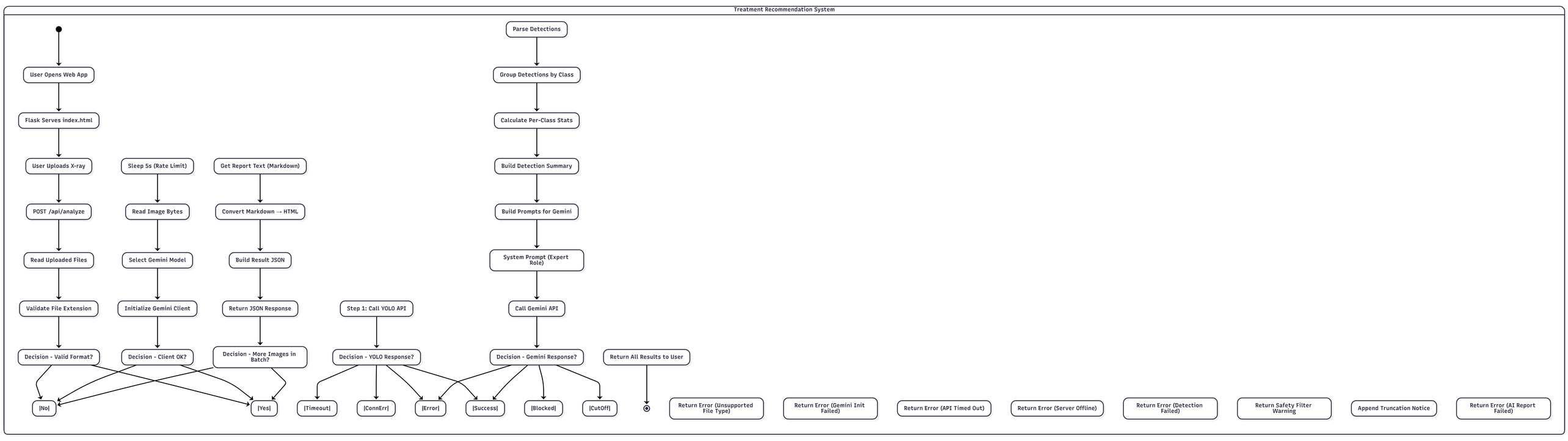
Treatment Recommendation System
User Opens Web App
Flask Serves index.html
User Uploads X-ray
POST /api/analyze
Read Uploaded Files
Validate File Extension
Decision - Valid Format?
|No|
|Yes|
Sleep 5s (Rate Limit)
Read Image Bytes
Select Gemini Model
Initialize Gemini Client
Decision - Client OK?
Get Report Text (Markdown)
Convert Markdown → HTML
Build Result JSON
Return JSON Response
Decision - More Images in Batch?
Step 1: Call YOLO API
Decision - YOLO Response?
|Timeout|
|ConnErr|
|Error|
|Success|
|Blocked|
|CutOff|
Parse Detections
Group Detections by Class
Calculate Per-Class Stats
Build Detection Summary
Build Prompts for Gemini
System Prompt (Expert Role)
Call Gemini API
Decision - Gemini Response?
Return All Results to User
Return Error (Unsupported File Type)
Return Error (Gemini Init Failed)
Return Error (API Timed Out)
Return Error (Server Offline)
Return Error (Detection Failed)
Return Safety Filter Warning
Append Truncation Notice
Return Error (AI Report Failed)